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Applied Analytics Project - Week 7 Assignment

Model Approach:

For Week 7, our group selected Random Forest Classification as our modeling approach. This decision aligns with our prediction objective of predicting NCAA tournament outcomes as the model uses a group of decision trees to make a final prediction. For our project, the target variable is game outcome, where Team 1 wins the matchup or loses.

Random Forest makes sense for this project because:

- There are a large number of numeric features.
- It is an ensemble model that averages predictions from multiple trees, instead of relying on one model.
- It works well for complex datasets where variables may not have linear relationships.

Model Complexity:

Random Forest models are more complex than other models, like our logistic regression model from last week. This is because it is an ensemble model that has many decision trees and aggregates their predictions. Instead of relying on a single decision tree, the Random Forest model uses a number of decision trees with random subsets of data and features to make independent predictions. The final prediction is then made using a majority vote across all of the decision trees.

Hyperparameters Evaluated:

We evaluated multiple hyperparameters to determine which configuration produced the best validation performance.

The main hyperparameters tested were:

- Number of Trees (n_estimators)
 - We tested different values to determine if increasing the number of trees improves the model.
- Maximum Tree Depth (max_depth)
 - We tested different values to evaluate how overfitting and underfitting presents in different depths.
- Minimum Samples Split (min_samples_split)
 - We tested different values to determine the minimum number of samples required before splitting a node.
- Maximum Features (max_features)
 - We tested different values to see how many features produced the best split.
- Class Weight (class_weight)
 - We tested none and balanced to adjust for class imbalance in the dataset.

In total, 72 model configurations were tested.

Model Performance Metrics:

To evaluate our Random Forest model, we used the following performance metrics:

- Accuracy
 - This gives a general understanding of how often the model correctly predicts the outcome of games.
- Precision
 - This is important because we want to limit false win predictions, especially since incorrect win predictions can significantly impact the outcome of brackets.
- Recall
 - This is important because we want the model to successfully capture as many true wins as possible rather than missing them.

Model Variations:

For this week, we evaluated 72 different Random Forest configurations. Each configuration was trained on seasons 2003-2021 and validated using the 2022 season. For every model, we calculated accuracy, precision, and recall on the training and validation datasets.

	configId	model	set	accuracy	precision	recall
0	0	{'classWeight': None, 'maxDepth': None, 'maxFe...	train	1.000000	1.000000	1.000
1	0	{'classWeight': None, 'maxDepth': None, 'maxFe...	val	0.730159	0.734694	0.900
2	1	{'classWeight': None, 'maxDepth': None, 'maxFe...	train	1.000000	1.000000	1.000
3	1	{'classWeight': None, 'maxDepth': None, 'maxFe...	val	0.746032	0.740000	0.925
4	2	{'classWeight': None, 'maxDepth': None, 'maxFe...	train	1.000000	1.000000	1.000
...	...	...	...	...	...	...
139	69	{'classWeight': 'balanced', 'maxDepth': 20, 'm...	val	0.730159	0.709091	0.975
140	70	{'classWeight': 'balanced', 'maxDepth': 20, 'm...	train	1.000000	1.000000	1.000
141	70	{'classWeight': 'balanced', 'maxDepth': 20, 'm...	val	0.761905	0.735849	0.975
142	71	{'classWeight': 'balanced', 'maxDepth': 20, 'm...	train	1.000000	1.000000	1.000
143	71	{'classWeight': 'balanced', 'maxDepth': 20, 'm...	val	0.746032	0.722222	0.975

We selected the top three best-performing models by ranking all configurations based on validation accuracy and choosing the three with the highest validation performance.

Discuss and Analyze how training and validation metrics change across variations:

Across the top three configurations (5, 16, 17), training performance was much higher than validation performance. This indicates that overfitting is present, which is expected in Random Forest models. Training accuracy was 0.99-1.00, while validation accuracy was 0.78-0.79 across the three top models. Training precision was 0.99-1.00 and validation precision was 0.75-0.76. Lastly, training recall was 1.00 and validation recall was 0.975 for all three model configurations. The differences in training and validation accuracy and precision demonstrates that there is overfitting present. However, the high validation recall shows that the models correctly identify most of the true winning teams.

Compare performance metrics for the validation dataset across all 3 variations:

The top three models had very similar validation performance. However, configuration 16 had slightly higher accuracy and precision than configurations 5 and 17. All three recalls are the same. Therefore, configuration 16 is the top model with the below validation results:

- Validation Accuracy: 0.794
- Validation Precision: 0.765
- Validation Recall: 0.975

	configId	model	set	accuracy	precision	recall
10	5	{'classWeight': None, 'maxDepth': None, 'maxFe...	train	0.999139	0.998726	1.000
11	5	{'classWeight': None, 'maxDepth': None, 'maxFe...	val	0.777778	0.750000	0.975
32	16	{'classWeight': None, 'maxDepth': 10, 'maxFeat...	train	0.991394	0.987406	1.000
33	16	{'classWeight': None, 'maxDepth': 10, 'maxFeat...	val	0.793651	0.764706	0.975
34	17	{'classWeight': None, 'maxDepth': 10, 'maxFeat...	train	0.990534	0.986164	1.000
35	17	{'classWeight': None, 'maxDepth': 10, 'maxFeat...	val	0.777778	0.750000	0.975

	configId	model	set	accuracy	precision	recall
33	16	{'classWeight': None, 'maxDepth': 10, 'maxFeat...	val	0.793651	0.764706	0.975
11	5	{'classWeight': None, 'maxDepth': None, 'maxFe...	val	0.777778	0.750000	0.975
35	17	{'classWeight': None, 'maxDepth': 10, 'maxFeat...	val	0.777778	0.750000	0.975

Even though configuration 16 is identified as the best performing configuration, the differences between the top three configurations is marginal.

Identify the best model for the week and discuss why you selected that model:

The best model selected for Week 7 was configuration 16 because it had the highest ranked accuracy and precision. In addition, its recall matched that of the other top two model configurations. By having the top accuracy and precision numbers and being tied for the top recall number, configuration 16 was the clear top ranked Random Forest model this week.